

SAS CU Hackathon

Let's Showcase The Skills !



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Problem Context Brief



The Problem Context

We provide Datasets & Context !

CONTEXT : BIG PICTURE



The problem is around data science related jobs, skills and personality traits of the data scientists.

You may think of myriad of analysis including data management, visualizations, patterns , statistical or data mining analysis with a definitive objective to bring out best information out of available data.

The Context

>> DATA POINTS : BIG PICTURE

1. Data Science Job Postings including descriptions, salary, experience and so on
2. Skill traits in data scientists for success including technical skills like coding, math, AI, ML and dashboards and story telling
3. Personality traits including neuroticism, extraversion, openness to experience, agreeableness and conscientiousness.

The Data Description doc (Refer to PDF in the system)

Data Description

There are four data files. Here is a brief description of the files.

File	Description
Data Science Jobs	Around 1600 rows of sample data describing the number of jobs posted by the leading companies including details like salary, experience and so on in the years 2024-25
Analytics Jobs	Around 15800 rows of sample data describing job postings in analytics area including details like job description, designation , location salary and so on in the years 2024-25
JDS Skill Traits (Junior Data Scientists)	Around 140 rows of sample data of a company's skillset data of its junior or entry level data scientists. The data presents measured technical skillsets and outcome like promotion or salary hike (high/low).
SDS Personality Traits (Senior Data Scientists)	Around 160 rows of sample data of a company trait analysis data of its senior and customer facing level data scientists. The data presents measured personality traits and outcome like overall success (high/low).

Data Dictionaries

File: *Data Science Jobs*

Column	Brief Description
reference_no	Row identification number or ID
company_name	Name of the recruiting company
job_title	Title of the job
min_experience	Minimum required experience
avg_salary	Average salary offered across the job postings for the company
min_salary	Minimum salary offered across the job postings for the company
max_salary	Maximum salary offered across the job postings for the company
num_of_jobs	Number of posting by the company

File: *Analytics Jobs*

Column	Brief Description
s_no	Row identification number or ID
experience	Years of experience required for the job
job_description	Typical description of the job
job_desig	Designation or position of the job role
job_type	Type or classification of the job
key_skills	Key skills required for the job
location	Geo location of the job requirement
salary	Salary offered

File: *JDS Skill Traits*

Column	Brief Description
id	Row identification number or ID
big_data_skills	Average score on data skills , summarized on a scale of 1-5 based on multiple tests, training cum evaluations and feedback.
maths-stats_skills	Average score on quantitative, maths and statistics skills, summarized on a scale of 1-5 based on multiple tests, training cum evaluations and

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maths-stats_skills	Average score on quantitative, maths and statistics skills, summarized on a scale of 1-5 based on multiple tests, training cum evaluations and feedback.
coding_skills	Average score on coding skills on SAS, Python, SQL and so on, summarized on a scale of 1-5 based on multiple tests, training cum evaluations and feedback.
ai_and_ml_skills	Average score on newer concepts of artificial intelligence and machine learning, summarized on a scale of 1-5 based on multiple tests, training cum evaluations and feedback.
dashboard_and_storytelling_skills	Average score on visualization, reporting and storytelling, summarized on a scale of 1-5 based on multiple tests, training cum evaluations and feedback.
salary_hike_high_or_low	A binary variable for performance-based salary hike 1= high, 0 = low

Junior Data Scientist Skill Sets

Average score on big data, quantitative, maths and statistics skills,AI, ML, visualization and so on summarized on a scale of 1to 5 based on multiple tests, training cum evaluations and feedback.

Data Dictionaries

- Senior and Customer facing data scientists
- The table presents the measurements on personality traits that exist on a spectrum, with individuals falling anywhere between high and low.
- The Big Five model, or Five-Factor Model (FFM), is a widely accepted psychological framework that categorizes human personality into five broad, stable dimensions:
- Openness, Conscientiousness, Extraversion, Agreeableness, and Neuroticism
- Often abbreviated as OCEAN or CANOE.

Column	Brief Description
id	Row identification number or ID
neuroticism	It measures the neuroticism which is a fundamental personality trait within the "Big Five" model characterized by a chronic tendency to experience negative emotions, such as anxiety, depression, anger, and self-doubt. The measurement is presented on a normalized score. Higher numbers represent high levels or values.
extraversion	It measures the extraversion which represents a spectrum of social engagement, ranging from high energy, outgoing, and assertive (extraverted) to quiet, reserved, and solitary (introverted) behaviours. The measurement is presented on a normalized score. Higher numbers represent high levels or values.
openness_to_experience	Openness to experience in the Big Five model measures a person's level of creativity, curiosity, and willingness to embrace new ideas, experiences, and unconventional thinking. The measurement is presented on a normalized score. Higher numbers represent high levels or values.
agreeableness	Agreeableness in the Big Five personality model measures an individual's tendency to be compassionate, cooperative, trusting, and empathetic rather than suspicious or antagonistic. The measurement is presented on a normalized score. Higher numbers represent high levels or values.
conscientiousness	Conscientiousness in the Big Five model represents a person's tendency to be organized, responsible, hardworking, and goal directed. The measurement is presented on a normalized score. Higher numbers represent high levels or values.
success_classification_high_low	It presents high and low classification of success levels within the organization.

Data : Key Considerations

1. This dataset contains public, self-reported or masked information.
2. May contain meaningless/misspelled/mistyped entries.
3. It may be present with outliers and other problems
4. The personality traits have been shown on normalized data
5. *You must upload the data to SAS VFL if you wish to use SAS Software.*

Time & Resource Management

1. Manage Time & Resource Judiciously
2. Divide tasks between team members
3. Practice Software components
4. Spend time on initial part i.e. formulating your problem statement

Separate Instructions to access the software shall be provided

What would be tested ??

*Note : Everything depends on your Problem Identification or Analytics Objective .
Be careful , thoughtful and Do not Haste . You may spend a good amount of time
here*

>> The problem identification or Analytics Objective should allow for testing of following items:

1. Data Management and Data Manipulation skills
2. Approach to solve problem in Business Analytics Context
3. Statistical Skills / Data Mining Skills/Machine learning skills
4. Overall conclusion drawn from the data analysis

Agenda

Expected Outcome Formats

Approach Note : Important for Round 2

Problem definition / Analytics Objective

Approach:

Data Exploration: data derivation,
data modification, Data Deduction, reductions

Data Analysis:

Results and Conclusions:

Implications:

Word Format

*20 – 25 pages excluding
Appendix*

*Time Roman , 12 , single
spacing preferred*

*May include Appendix ,
which may have :*

*Tables, Graphs , Codes,
Software screenshots,
Equations etc. and any
other info*

Approach Note : Round 2 Considerations

Total – 100 Marks (Top 8 Teams)

Main Headings	Marks	Key Considerations - A generic guideline
Problem definition or Analytics Objective	10	Problem Identification Skills & Scope, Depth & Coverage of Identified Problem
Approach Description	15	Describe the overall flow , motivations and reasons to adopt the approach
Data Exploration: (Data Manipulation, Data Derivation, Consolidation, Preparation etc.)	25	Skills to identify data issues , approach to solve issues , consolidate information out of data, exploratory skills and data preparation strategies
Data Analysis	30	Skills to analyse the data , statistical / non statistical skills, descriptive / prescriptive analytical skills, Overall Approach
Results and Conclusions	10	Ability to consolidate information from the previous data analysis , linkage to the problem statement and solution description
Implications	10	implications of the finding to the relevant stakeholders in society or stakeholders concerned

Presentation – Round 3 (Top 8 teams will present)

Power Point Format

15 – 25 Slides

*Decide your own
template, format and
design*

*Ten minutes to present
max*

3-5 mins for Jury Q & A

*May include VA based
Graphics*

For the 10 Teams chosen in Round 2

1. Presentation and communication skills
2. Usage of graphics ,
3. Slide Management
4. Overall Presentation Storyline
5. Concluding slides
6. Q & A by Jury

Duration and flow shall be notified later

Round – 3, Presentation – 100 marks

Main Headings	Marks	Genric Considerations
Presentation and communication skills	25	body langague, confidence , langague , eye contact & connect with the audience
Usage of graphics	10	Usage of graphs , visuals etc. to prove or disapporve a point
Slide Management	5	Slide sequencing , information distribution , time distribution
Overall Presentation Storyline	20	Ability to create a logical connect from statement to the conclusion
Concluding slides	15	Concluding remarks , emphatic and definitive comments
Q & A by Jury	25	Ability to handle questions , alertness & assertiveness

Winner

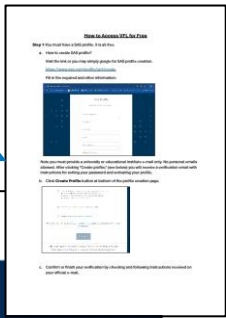
Round 2 Marks (70 %)+ Round 3 Marks (30 %) = Total (Top 3)

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Technology Options

VFL for SAS Users

Click to open the pdf



Software Learn Support Partners About Us SAS Innovate



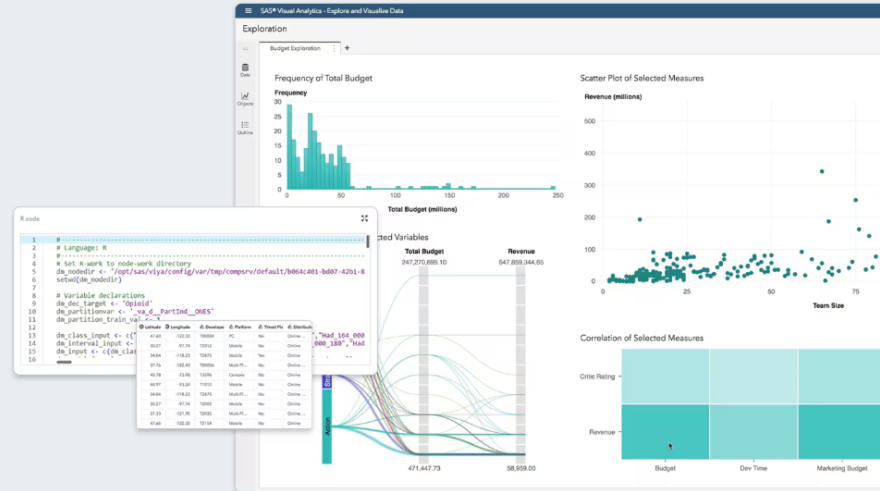
SAS* Viya* for Learners

Advance your expertise in cutting-edge data science skills

To access this product, please sign in with your university email.

Access for educators

Access for students



You Need SAS Profile and University Mail ID for SAS VFL

Technology Agnostic



Load data to VFL



Make a Report in SAS Visual Analytics



Make a Model in SAS Model Studio



Do's & Don'ts

Important Note - Do's & Don'ts

1. Excel may be used only for preliminary analysis but discouraged
2. No e-mail access or data transfers to outside world during night or day
3. Unfair means , indiscipline , non-cooperation , insults to other teams etc. shall lead to disqualification
4. Cross-team discussions not encouraged
5. Any damage to systems or software not to be tolerated
6. No chance to influencing the mentors or jury members inside or outside premises
7. Option to quit before time though not encouraged or suggested
8. Decision of jury to be final and binding , no appeals or repeals .
9. Manage Time & Resource Judiciously
10. Divide tasks between team members
11. Practice Software components
12. Spend time on initial part i.e. formulating your problem statement

All the Best !!



*all the
best*